

AMAZON MUSIC CLUSTERING – Final Report

SUMMARY

This project uses **K-Means clustering** to group Amazon Music tracks based on audio features. After preprocessing and feature scaling, three distinct clusters were identified and analyzed using bar charts, heat maps, and scatter plots. The clusters represent **chill acoustic tracks**, **high-energy party hits**, and **speech-heavy experimental songs**, demonstrating how unsupervised learning can uncover meaningful music patterns.

THE OBJECTIVE:

- Group songs into meaningful categories using **unsupervised learning**
 - Identify patterns in music based on **audio features**
 - Interpret and label each group in a musically meaningful way
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APPROACH

Data Preparation

- Selected relevant numerical audio features such as:
danceability, energy, acousticness, speechiness, loudness, tempo, valence, etc.
- Removed non-informative or categorical columns
- Applied feature scaling to ensure equal contribution of all features

Clustering Technique

- Used K-Means clustering, an unsupervised machine learning algorithm
- Determined the optimal number of clusters as $K = 3$ using the methods such as
 1. **Silhouette Score:** Measures how close points are to their own cluster vs. others.
 2. **Davies-Bouldin Index:** Lower value means better separation.
 3. **Inertia (for K-Means):** Measures cluster compactness.
 4. **Elbow Method:** Plot SSE (inertia) vs. k and find the "elbow."
- Assigned each song to one of the clusters: 0, 1, or 2

Cluster Profiling

- Calculated mean values of audio features per cluster
- Interpreted clusters based on dominant musical characteristics

Visualizations Used

To understand and validate the clustering results, multiple visualization techniques were applied:

Bar Charts

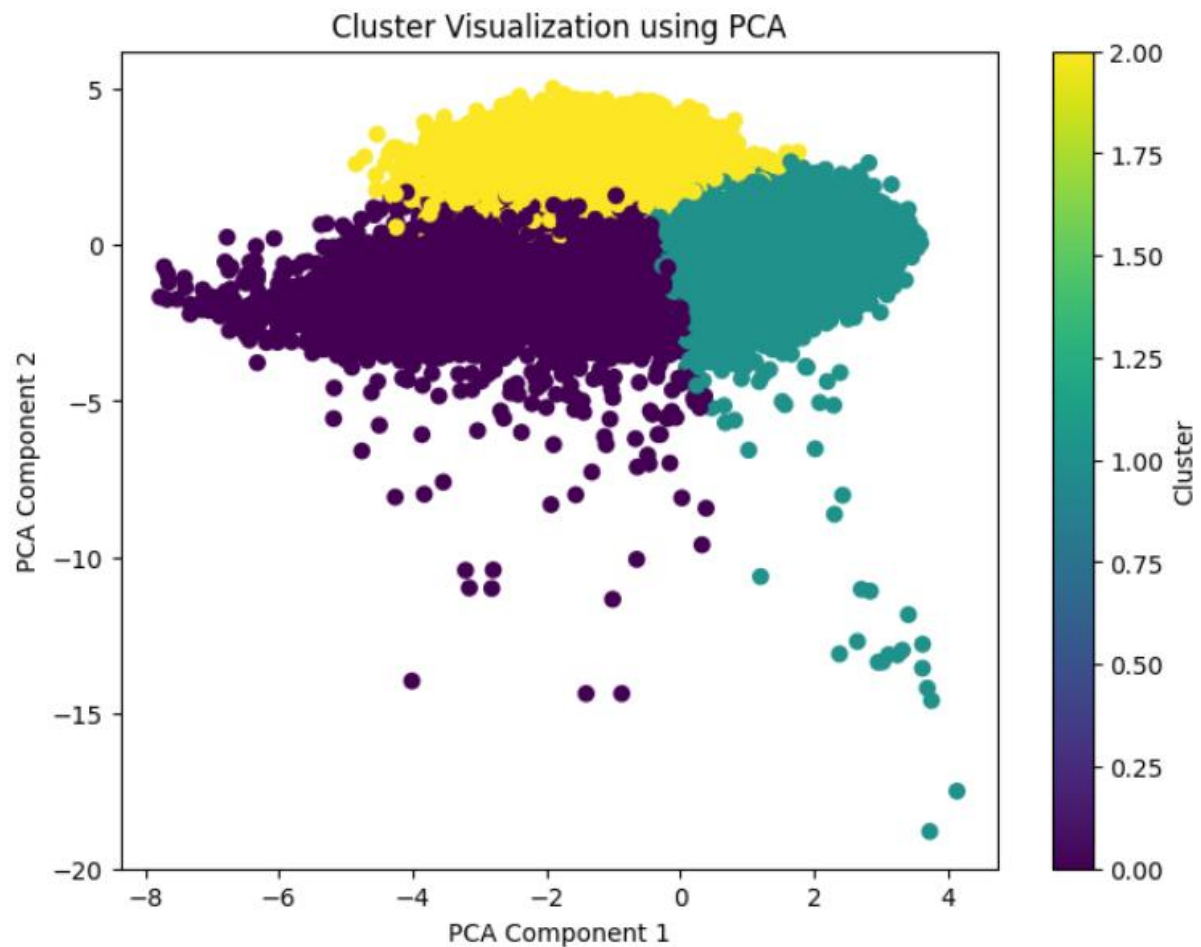
- Showed average feature values per cluster
- Helped compare clusters feature-by-feature (e.g., energy, acousticness)

Heat maps

- Used normalized feature values
- Provided a **holistic comparison** of all clusters at once
- Clearly highlighted dominant traits of each cluster

PCA Scatter Plot

- Reduced high-dimensional data to 2 components
- Enabled clear visualization of cluster separation in 2D space



CLUSTER ANALYSIS & INTERPRETATION

Cluster 0 – “Chill Acoustic / Soft Listening Tracks”

Key statistics

- Danceability: 0.49 (moderate)
- Energy: 0.31 (low)
- Loudness: -13.2 dB (soft)
- Acousticness: 0.75 (very high)
- Instrumentalness: 0.17 (some instrumental content)
- Tempo: 112 BPM (moderate)
- Valence: 0.41 (slightly calm / neutral mood)
- Duration: ~223 sec (~3.7 min)

Musical interpretation

Songs in this cluster are:

- Soft and calm
- Heavily acoustic
- Not aggressive or loud
- Suitable for relaxed listening

They are not designed for dancing, but they flow nicely for:

- Background music
- Studying
- Evening listening
- Acoustic playlists

Final label: “Chill Acoustic / Soft Listening Tracks”

Business insight

These tracks fit well in:

- “Relax & Unwind” playlists
 - Café / ambient music
 - Focus & calm listening experiences
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Cluster 1 – “High-Energy Party / Mainstream Tracks”

Key statistics

- Danceability: 0.63 (high)
- Energy: 0.69 (high)
- Loudness: -7.6 dB (very loud)
- Acousticness: 0.26 (low)
- Instrumentalness: 0.05 (mostly vocals)
- Tempo: 125 BPM (fast)
- Valence: 0.67 (happy / positive)
- Duration: ~226 sec (~3.8 min)

Musical interpretation

This cluster clearly represents:

- Upbeat
- Loud
- Energetic
- Dance-friendly songs

These are modern, commercial, mainstream tracks that:

- Work well in parties
- Are popular on streaming platforms

- Have strong vocal presence

Final label: “Party / Mainstream Hit Tracks”

Business insight

Ideal for:

- Workout playlists
 - Party & club music
 - Trending / viral content
 - Radio & mass-appeal audiences
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Cluster 2 – “Experimental / Speech-Heavy or Short-Form Tracks”

Key statistics

- Danceability: 0.66 (high)
- Energy: 0.47 (moderate)
- Loudness: -13.4 dB (soft)
- Speechiness: 0.83 (very high)
- Acousticness: 0.59 (moderate-high)
- Instrumentalness: ~0.00 (vocals dominant)
- Tempo: 100 BPM (slow-moderate)
- Valence: 0.58 (neutral-positive)
- Duration: ~98 sec (~1.6 min) ← very short

Musical interpretation

This cluster is very distinctive:

- Extremely high speechiness → spoken-word, rap-like, narration
- Short duration → snippets, intros, skits, viral audio

- Moderate energy → not aggressive music
- Balanced mood

These could be:

- Spoken poetry
- Interludes
- Storytelling tracks
- Viral short-form audio

Final label : “Speech-Heavy / Short-Form Experimental Tracks”

Business insight

Best suited for:

- Podcasts or spoken-word content
 - Storytelling music
 - Experimental / niche audiences
 - Social media or short-attention formats
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CONCLUSION

This project demonstrates how **unsupervised learning** can be effectively used to analyze and categorize music based on audio features. By combining clustering with strong visualization and interpretation, raw numerical data was transformed into meaningful and actionable musical insights.